

MOHAMED MARWEN MAALAWI

Senior Full-Stack Software Engineer

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PROFESSIONAL SUMMARY

Senior Full-Stack Software Engineer with a track record of architecting, building and shipping production-grade platforms from zero to scale. I own the full technical lifecycle — from system design and data modeling through API architecture, CI/CD pipelines and production monitoring. Core expertise in NestJS, Spring Boot, Next.js and React Native. Proven ability to lead technical decisions on distributed systems, financial integrations (Stripe, QuickBooks), workflow engines (Camunda BPM) and AI-powered features. I build systems designed to scale without architectural rewrites, and I deliver with full autonomy in fast-paced, international engineering teams.

PROFESSIONAL EXPERIENCE

NUMERYX TUNISIA

June 2025 – Present

Software Engineer

- Architected and implemented Spring Boot microservices handling distributed business workflows with fault-tolerant inter-service communication
- Designed and deployed Camunda BPM process definitions for multi-step business automation, reducing manual coordination overhead by an order of magnitude
- Engineered a GraalVM-based polyglot script execution sandbox, enabling dynamic runtime script evaluation within workflow tasks without JVM restarts
- Built production-hardened error recovery mechanisms — circuit breakers, retry policies and dead-letter queues — ensuring zero silent failures in critical business flows
- Integrated Keycloak for enterprise-grade identity management: RBAC, SSO, token-based auth and fine-grained permission policies across all services
- Led architecture discussions and drove technical decision-making for distributed system design, defining service boundaries, data ownership and API contracts

Technologies: Spring Boot • Java • Camunda BPM • GraalVM • Keycloak • Docker • Microservices

LUDDITECH INC

January 2023 – April 2025

Full-Stack Engineer

- Owned the full-stack architecture of multiple SaaS products and React Native mobile applications serving international B2B clients across North America
- Designed and built real-time communication infrastructure — WebSocket-based chat, presence systems and cross-platform push notification pipelines via Firebase and APNs
- Architected a comprehensive Stripe integration: subscription lifecycle management, idempotent webhook processing, automated dunning flows, billing portal and complete audit logging — zero billing incidents post-launch
- Engineered a bi-directional QuickBooks sync engine with conflict resolution, exponential backoff retry, dead-letter queues and automated reconciliation — eliminating 100% of manual accounting effort
- Designed clean AI integration layers with provider abstraction, streaming response support, per-tenant cost tracking and graceful fallback chains — enabling rapid model switching without code changes
- Established CI/CD pipelines with GitHub Actions — automated testing, linting, staging deployments and production release gates, reducing deployment time from hours to minutes
- Operated as a key technical decision-maker in an international remote team, independently owning features from RFC through production deployment and monitoring

Technologies: NestJS • React Native • Node.js • Stripe • QuickBooks • Firebase • OpenAI • GitHub Actions

SAVY

September 2021 – December 2022

Lead Mobile Engineer

- Architected a complex React Native mobile application for UK tax recovery, enabling British citizens to reclaim tax overpayments via automated HMRC self-assessments
- Engineered dynamic, multi-step tax deduction forms using React Hook Form and custom Bottom Sheet architectures for efficient receipt categorization and OCR parsing
- Implemented a robust offline-first synchronization engine using WatermelonDB (SQLite over JSI), ensuring 60 FPS scrolling and immediate data availability without network connectivity
- Optimized list rendering and memory footprint using FlashList and Hermes engine, eliminating layout thrashing and JS frame drops on legacy iOS/Android devices
- Integrated complex functional UI components—segmented controls, interactive tax return status bars, and custom date pickers—prioritizing usability and data density
- Automated the mobile CI/CD pipeline with Fastlane, enabling predictable App Store/Google Play deployments and Over-The-Air (OTA) updates via CodePush

Technologies: React Native • TypeScript • WatermelonDB (JSI) • Hermes • CodePush • Fastlane • Redux

FIZL

March 2020 – August 2021

Full-Stack Software Engineer

- Led the architectural design and development of Fizl's core tax management and accounting engine for independent contractors and SMEs
- Engineered automated ledger reconciliation algorithms and tax provision modules, reducing manual accounting errors by over 90%
- Developed comprehensive financial dashboards using Next.js and Tailwind, visualizing real-time cash flow, pending liabilities, and deductible business expenses
- Built a highly scalable RESTful backend in Node.js/NestJS capable of ingesting and categorizing thousands of banking transactions daily via Open Banking APIs
- Designed database schemas in PostgreSQL to safely store immutable financial records, ensuring strict compliance with financial auditing standards

Technologies: NestJS • Next.js • PostgreSQL • Open Banking • AWS • TypeScript

Technical Lead & Full-Stack Architect

- Led end-to-end product architecture and development for clients in e-commerce, logistics and fintech — from initial technical scoping to production launch
- Designed NestJS backend architectures with PostgreSQL, implementing domain-driven design, modular service boundaries and scalable REST/GraphQL API layers
- Built production Next.js frontends with ISR, server components, optimized Core Web Vitals and comprehensive SEO strategies
- Delivered cross-platform React Native mobile applications with offline-first architecture, push notifications and native module integration
- Established containerized deployment pipelines with Docker Compose, multi-stage builds and environment-specific configuration management
- Provided technical consulting and architecture reviews for product teams — database schema design, API contract definition and technology selection

Technologies: NestJS • Next.js • React • React Native • PostgreSQL • Docker • TypeScript

TECHNICAL SKILLS

BACKEND & ARCHITECTURE

NestJS • Spring Boot • Node.js • Java • TypeScript

DATABASES

PostgreSQL • MySQL • MongoDB • NoSQL • Redis

TOOLS & METHODOLOGIES

Stripe • QuickBooks • Keycloak • Camunda BPM • GraalVM • Meta Pixel • Google Analytics

FRONTEND & MOBILE

Next.js • React • Angular • TypeScript • TailwindCSS • React Native • Expo

CLOUD & DEVOPS

Docker • GitHub Actions • CI/CD • Vercel